

Networks

Capacities

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Optimization: principles and algorithms



Capacities

There is a limit to the quantity of flow that can be transported on each section of a physical network

For instance, the quantity of water flowing through a pipe depends on the section area

One lane of a highway cannot accommodate more than 2400 veh/h

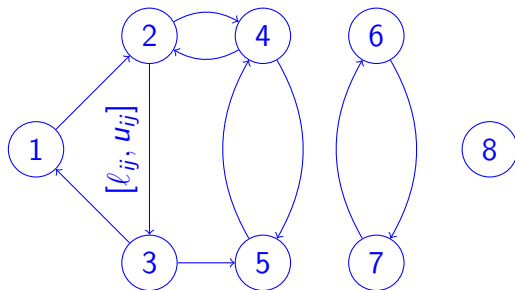
This limit is called the capacity.

In our mathematical formalism, we may impose bounds on flows.

Definition

$$\ell_{ij} \leq x_{ij} \leq u_{ij}.$$

- ▶ $\ell_{ij} \in \mathbb{R}$: minimum quantity of flow.
- ▶ $u_{ij} \in \mathbb{R}$: maximum quantity of flow.
- ▶ Associated with each arc (i, j) .
- ▶ Units are the same as x_{ij} .
- ▶ In practice, we often have:
 - ▶ $\ell_{ij} = 0$, or,
 - ▶ $\ell_{ij} = -u_{ij}$.



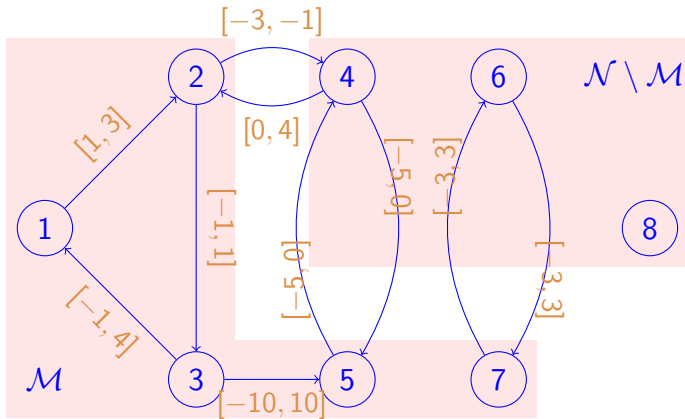
Capacity of a cut

$$U(\Gamma) = \sum_{(i,j) \in \Gamma \rightarrow} u_{ij} - \sum_{(i,j) \in \Gamma \leftarrow} \ell_{ij}.$$

$$\begin{aligned} \sum_{(i,j) \in \Gamma \rightarrow} u_{ij} &= \\ u_{24} + u_{54} + u_{76} &= -1 + 0 + 3 = 2 \end{aligned}$$

$$\begin{aligned} \sum_{(i,j) \in \Gamma \leftarrow} \ell_{ij} &= \\ \ell_{42} + \ell_{45} + \ell_{67} &= 0 - 5 - 3 = -8 \end{aligned}$$

$$U(\Gamma) = 2 - (-8) = 10$$



Capacity of a cut

Upper bound on the flow

$$X(\Gamma) \leq U(\Gamma).$$

Saturated cut

$$X(\Gamma) = U(\Gamma).$$

Summary

Bounds can be imposed on flows.

They are called capacities.

They are defined for each arc.

Capacity of cuts can be derived from arc capacities.